PM2.5 forecast for the next 3 days – 36 cities

Bubbles and cells are coloured by the US EPA category for tomorrow.
Updated every time the forecast runs.

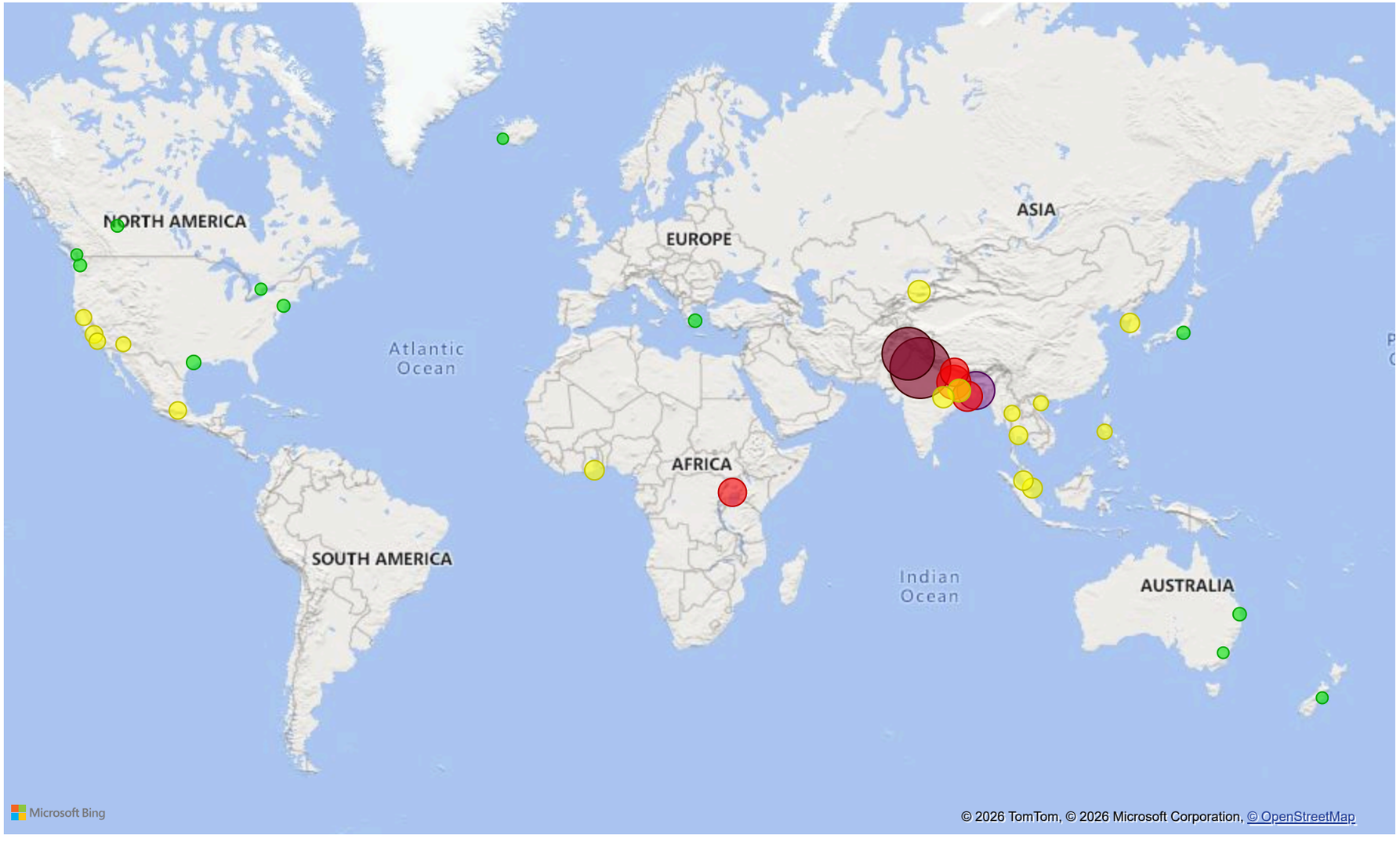
Forecast made

08 Nov 2026, 06:00 UTC

Cities unhealthy tomorrow

7

Tomorrow's forecast



name	24	48	72
Accra	23.40	22.30	21.00
Almaty	33.40	39.90	16.70
Athens	6.60	9.70	10.30
Bakersfield	17.20	16.60	17.10
Bangkok	19.80	20.70	21.00
Brisbane	6.70	7.20	7.90
Canberra	3.90	4.90	5.80
Chiang Mai	12.10	10.00	9.20
Christchurch	4.20	6.40	2.80
Delhi	390.30	349.20	315.20
Dhaka	130.00	103.40	76.10
Dhanbad	35.40	35.60	33.20
Edmonton	5.30	8.20	9.10
Hanoi	10.10	17.00	24.80
Houston	8.70	9.20	9.60
Kampala	63.00	71.80	75.80
Kathmandu	63.60	60.40	62.50
Kolkata	74.80	67.10	68.40
Korba	29.50	26.90	26.40
Kuala Lumpur	22.70	22.90	21.80
Lahore	281.60	301.00	284.80
Los Angeles	13.20	14.10	14.70
Manila	10.30	10.80	11.80
Mexico City	16.30	15.40	16.80
New York	5.60	4.40	6.80
Patna	100.70	98.10	92.20
Phoenix	9.80	10.40	9.80
Reykjavik	3.30	4.00	3.70

City forecast: the past week, the next 3 days, and why

Measured PM2.5 against this model, CAMS and persistence, with the drivers behind tomorrow's forecast.

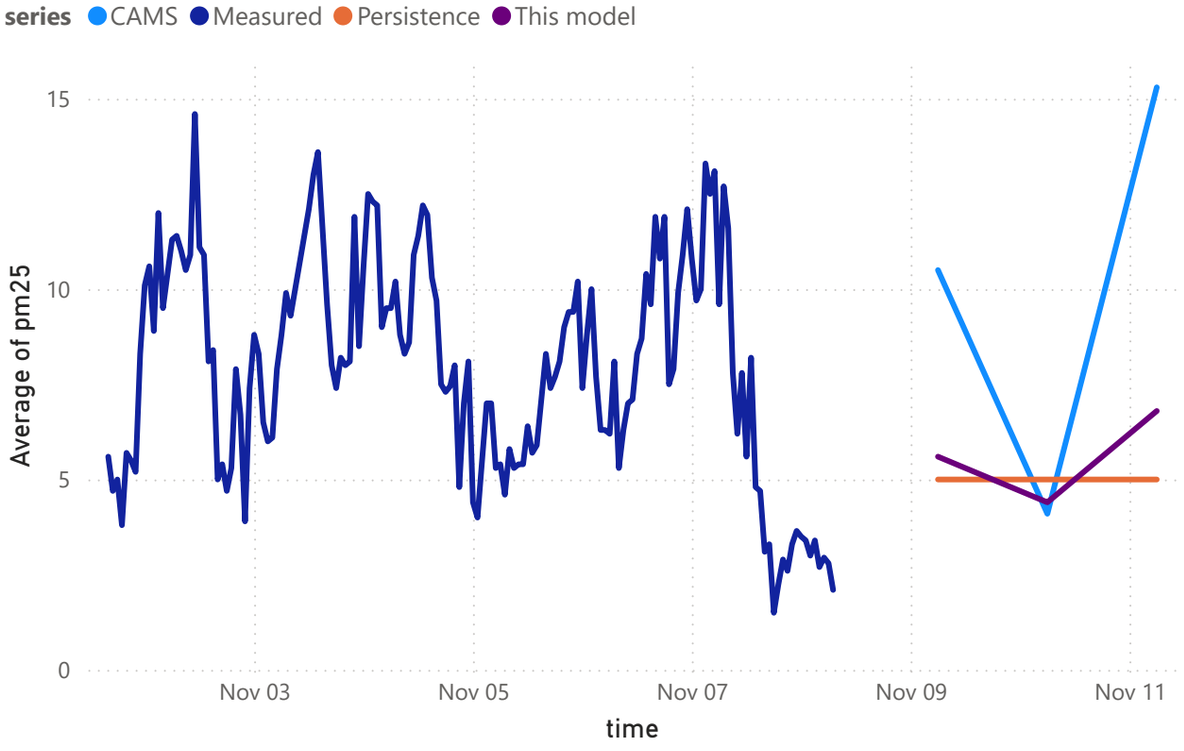
City Selection

New York

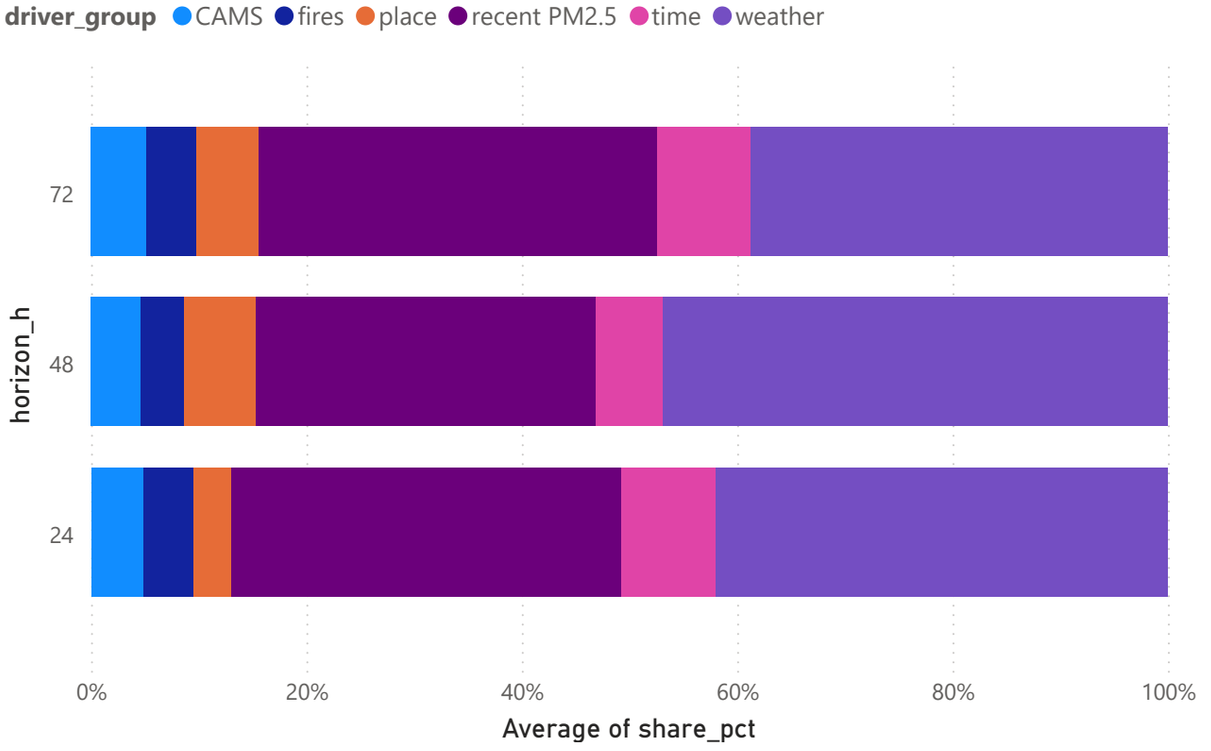
Top Feature	Effect on forecast (%)
wx_temp	8.50
wx_target_temp	-14.70
pm_72h	-15.30
wx_target_wind	-17.30
pm_3h	-27.00

+24h	+48h	+72h
6	4	7

Measured PM2.5 and the forecast (µg/m³)



What drives the forecast



How much better than “tomorrow looks like today”?

LightGBM tested on months it never saw: errors 27% smaller than persistence overall, and fires matter most on each city’s worst days.

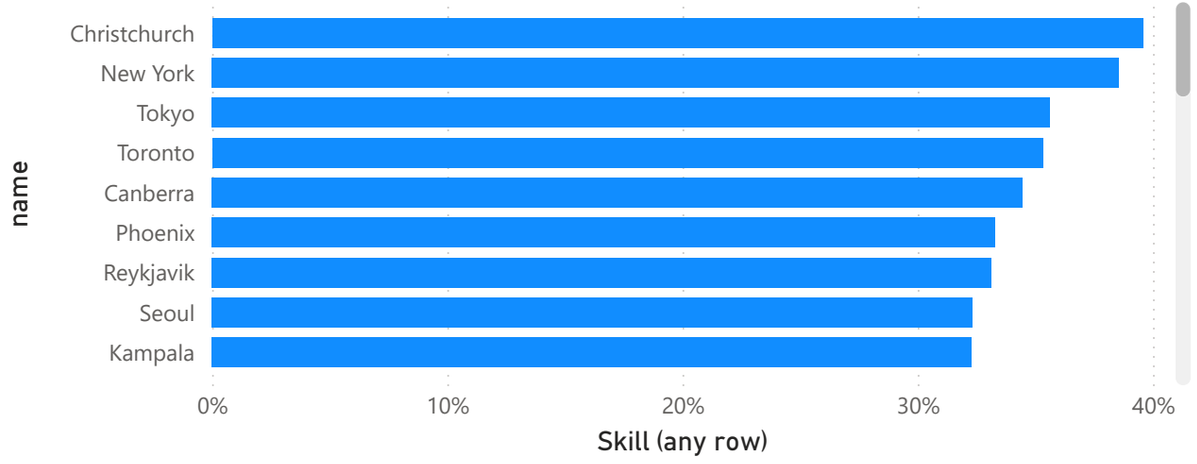
Error reduction vs persistence – held-out months

26.63%

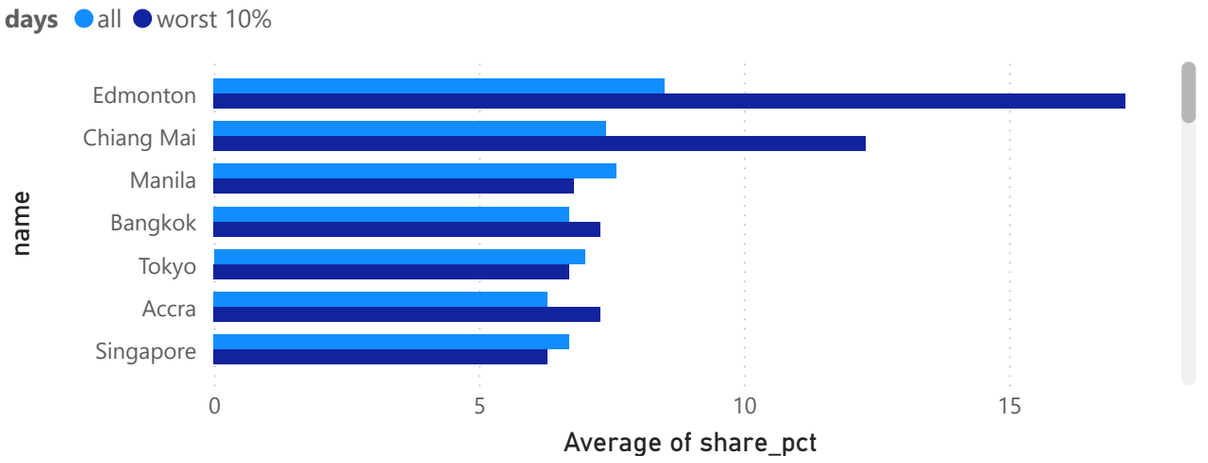
Error reduction vs persistence – most recent 90 days

16.80%

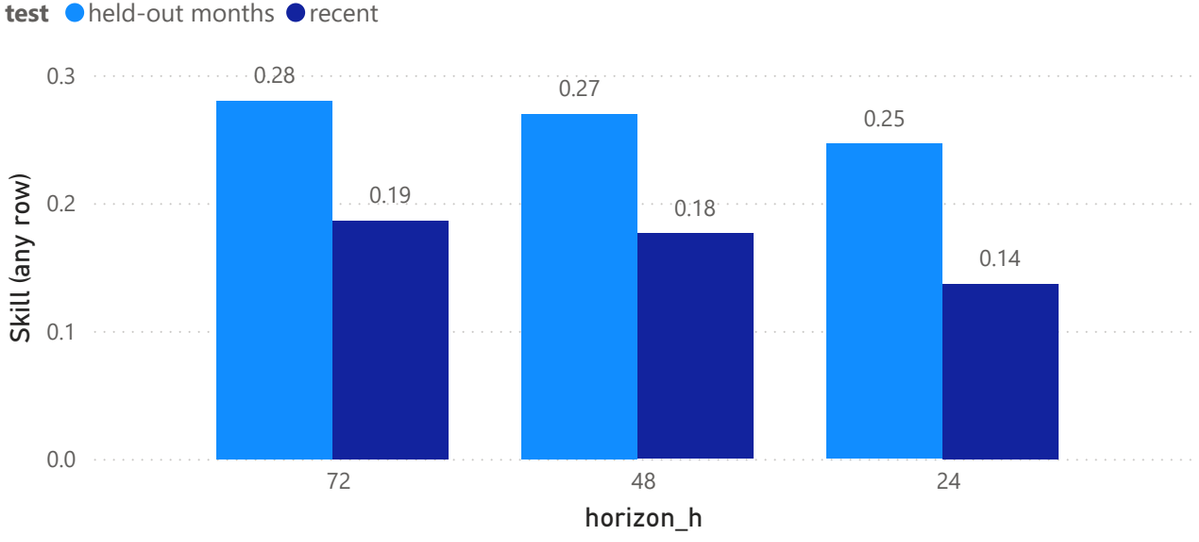
Error reduction vs persistence, by city



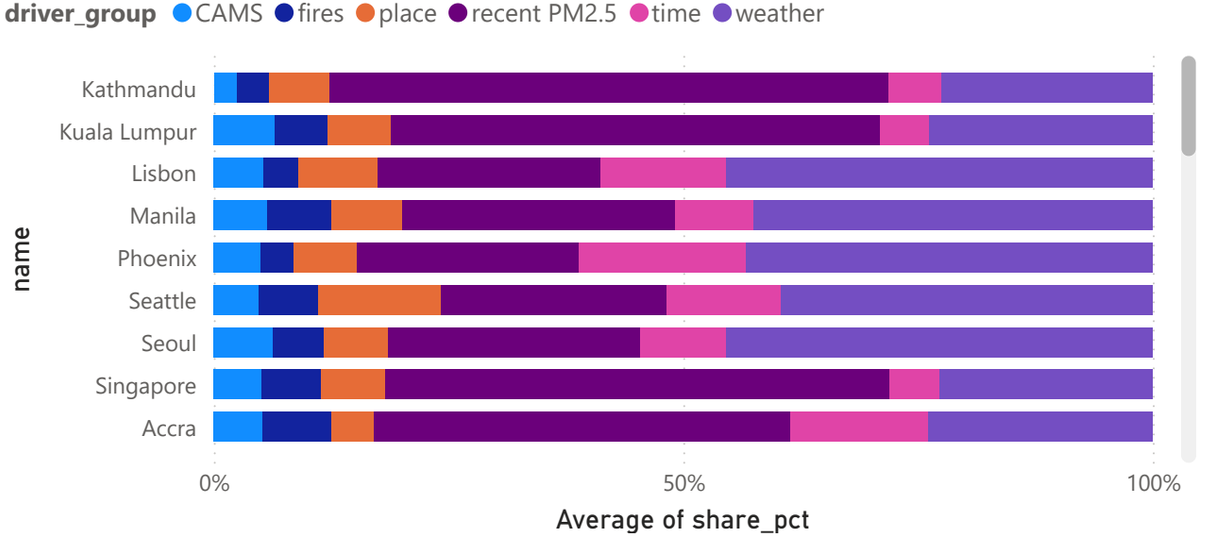
Share of the forecast explained by fires: all days vs each city’s worst 10%



Skill (any row) by horizon_h and test



Average of share_pct by name and driver_group



Live forecasts checked against what actually happened

Every saved forecast scored once its target time has passed, compared with CAMS and persistence. Grows daily.

Forecasts scored

266

Better than persistence

28.98%

Model error (µg/m³)

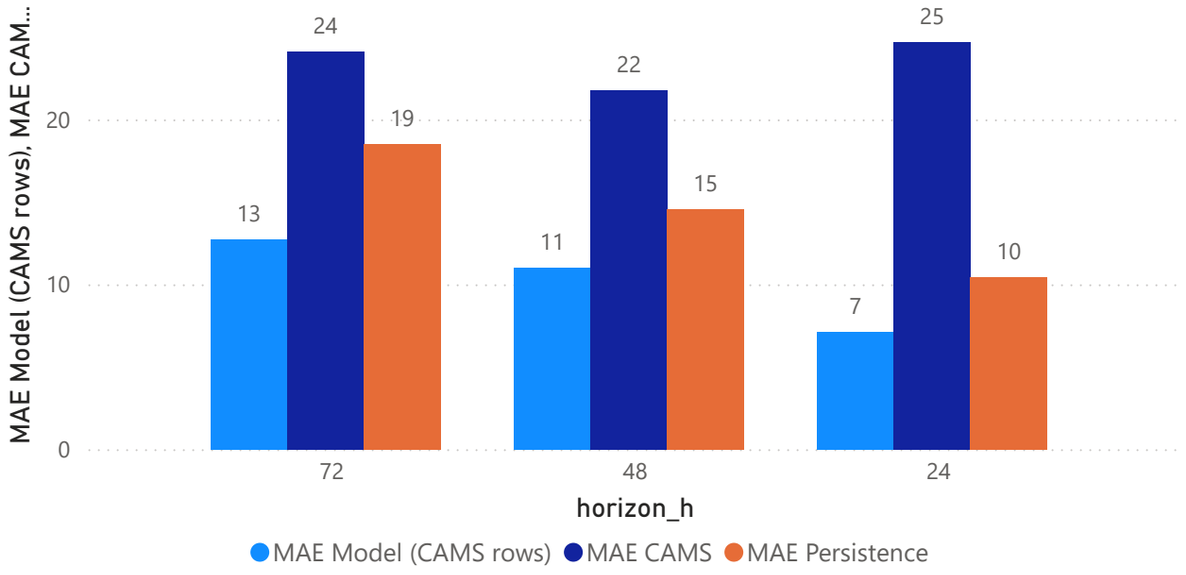
10.0

Better than CAMS

57.49%

name	row_id	issue_utc	horizon_h	pm25_adjusted	cams	persistence	Actual	Error
Accra	91	10/31/2026 6:00:00 AM	24	27.60	10.40	33.90	26.33	1.27
Accra	92	10/31/2026 6:00:00 AM	48	29.80	8.30	33.90	17.78	12.02
Accra	93	10/31/2026 6:00:00 AM	72	29.40	9.90	33.90	20.20	9.20
Accra	193	11/3/2026 6:00:00 AM	24	24.30	6.00	24.60	30.08	-5.78
Accra	194	11/3/2026 6:00:00 AM	48	22.40	5.00	24.60	15.04	7.36
Accra	195	11/3/2026 6:00:00 AM	72	23.20	5.40	24.60	15.50	7.70

Average error (µg/m³), lower is better



Forecast vs measured (one dot per forecast)

